

might exhibit different sound signatures. Let your sound preferences guide your choice, recognizing that everyone has different tastes.

## The role of CODECs

When reviewing headphone specifications, you'll encounter CODECs determining signal quality during Bluetooth audio transmission. Various CODECs exist, with SBC being the most common – basic and energy-efficient but compromising on audio quality. In noisy environments or with poorly tuned headphones lacking noise cancellation, the quality difference may be less noticeable. Consider factors such as streaming service quality, smartphone capabilities, and your headphone or earphone performance. For wired audio, the capabilities of your device's output port are crucial. Modern mobile devices without a headphone jack may need an external USB DAC for lossless audio. Additional hardware considerations apply to PCs. Here are brief descriptions of some popular CODECs found in headphones and earphones to navigate you through the audio world. 

## BUYING ADVICE

There were a few questions about buying headphones and earphones that came in the last few months. I turned to Satvik, our resident audio guy, to answer them. Here's a quick look –

### What TWS would you guys suggest around 2-2.5k?

 The first option for you would be the Realme Buds Air 3. They are slightly higher than your budget but are a great package overall. If you're looking for options strictly within your budget, you might want to consider the OnePlus Nord Buds 2 and the OnePlus Nord Buds 2r. They offer a bass-heavy sound, are feature-packed, and are great all-rounders.

Another option worth considering is the Oppo Enco Air3. It boasts a relatively balanced sound signature, appealing build and

design, and USB-C charging with decent battery life. If you are on an ultra-budget, the Redmi Buds 4 Active could be a good choice. They deliver decent performance in terms of sound, along with great build and fit for the price.

If you prefer something with a fairly neutral sound, then Moondrop and KZ are brands worth exploring. You might want to check out the Moondrop Chu or Chu 2, as well as the KZ ZSN Pro.

In case you have a bit more budget or are willing to save up, the Oppo Enco Air3 Pro stands out as your best bet. It's considered hands down the best pair of TWS earbuds you can get under 5K. Offering clean and detailed sound, fully customisable controls, support for LDAC codec, and a compact charging case, it ticks all the boxes for a premium experience.

| Codec   | Supported Bit Rates           | Special Features                                                 | Downsides                                                     | High-Res Audio Support |
|---------|-------------------------------|------------------------------------------------------------------|---------------------------------------------------------------|------------------------|
| SBC     | up to 345kbps (theoretically) | Low power consumption<br>Low processing power required           | Latency Issues<br>Compressed audio                            | No                     |
| AAC     | up to 320kbps                 | Better compression than SBC                                      | Not in the open domain<br>Uses more power and processing      | No                     |
| aptX    | up to 384kbps                 | Better compression algorithm than SBC                            | Not supported on iOS                                          | Yes                    |
| aptX HD | up to 576kbps                 | Improvement over aptX<br>Higher bit rate and bit depth than aptX | Not supported on iOS                                          | Yes                    |
| aptX LL | up to 352kbps                 | Low latency (about 32ms)                                         | Not supported on smartphones                                  | No                     |
| LDAC    | up to 990kbps                 | Supports High-Res Audio<br>Better support for Android devices    | Proprietary Sony technology<br>Limited presence in the market | Yes                    |
| ALAC    | approx. 1411kbps              | Supports lossless audio                                          | Limited to Apple devices                                      | Yes                    |